KRUPA PATEL

Structural Assessment of MAP4K4 and α-Synuclein as Parkinson's Disease Targets

Summary

This report evaluates MAP4K4 and α-synuclein (SNCA) as potential Parkinson's disease targets using predicted structural confidence, comparisons with experimental structures, pocket detection, and evidence from Modules 2 and 3. MAP4K4 has locally interpretable structural regions but substantial uncertainty in the arrangement of its domains. SNCA has a comparatively confident N-terminal region, while its C-terminal region is less certain and its conformation depends on its molecular environment. The pocket analysis does not, by itself, establish a druggable site for either predicted model. The final recommendations therefore distinguish biological target prioritization from evidence that a particular small-molecule binding site is structurally reliable.

Methods

AlphaFold Database structures were assessed using per-residue pLDDT confidence scores, predicted aligned error (PAE) maps, domain predictions, and UniProt annotations (Jumper et al., 2021; Varadi et al., 2022). Predicted structures were compared with experimental structures using sequence-paired superposition and per-residue deviation analysis. For MAP4K4, correcting the alignment procedure to semi-global alignment reduced the RMSD from 20.99 Å to 1.19 Å over 259 matched residues. Because fpocket could not be installed directly in the Windows environment, it was installed using micromamba in Google Colab. A guard was added to handle the relevant workflow condition, and the Step 6 cell was edited to read precomputed output. Additional checks included per-helix RMSD calculations for SNCA and inspection of residues contributing to structural deviations and predicted pocket boundaries. Pocket geometry was evaluated using fpocket, with confidence and biological context considered alongside the geometric scores (Le Guilloux et al., 2009). These computational measures were treated as evidence for prioritization, not as direct proof of biological function or druggability.

MAP4K4 has a mean pLDDT of 65.5, with 36.6% of residues below 50. The reported low-confidence stretches are residues 1–11, 174–189, 319–350, 526–539, and 550–912. The domain finder identifies regions 14–313, 354–522, and 917–1231. UniProt annotations place the MAP4K4 protein kinase domain at approximately residues 8–289 and the CNH domain at approximately residues 921–1219 (UniProt Consortium, n.d.). The first predicted region, 14–313, therefore broadly overlaps the kinase domain, although it extends beyond the annotated kinase boundaries. The third predicted region, 917–1231, closely corresponds to the annotated CNH domain. In contrast, the 354–522 region does not correspond to either of these two annotated domains and should not automatically be interpreted as a well-established folded domain.

The PAE map further limits how these regions can be interpreted. The three reported inter-domain PAE values are 28.1, 25.4, and 28.7 Å, indicating substantial uncertainty in relative domain placement. The region spanning residues 354–522 produces a thin diagonal feature rather than a distinct low-PAE square. Accordingly, the map does not provide strong support for treating this region as an independently positioned, rigid domain. The long low-confidence segment from residues 550–912 should likewise not be interpreted as a well-defined folded structure based on this prediction. Following the Module 4 slide 12 decision table, local regions with adequate confidence may be interpreted individually, but the relative arrangement of regions should not be considered reliable when inter-domain PAE is high (BIOT6900, Module 4 lecture slides, slide 12).

SNCA has a mean pLDDT of 75.2. Residues 1–88 are comparatively confident, whereas residues 89–140 have lower confidence, including residues 98–114. The domain finder identifies residues 1–88 as one domain, but the PAE map lacks a distinct low-PAE square supporting a rigid, independently positioned domain. For biological interpretation, the commonly used approximate regions are the N-terminal repeat region (residues 1–60), the NAC region associated with aggregation (residues 61–95), and the acidic C-terminal tail (residues 96–140). These sequence-region boundaries provide biological context rather than proving that each region is a separate folded domain (UniProt Consortium, n.d.). The comparatively high confidence in the N-terminal portion should not be taken as evidence that SNCA adopts one stable conformation in every environment. Ulmer et al. (2005) characterized the two-helical structure of micelle-bound α-synuclein, illustrating how lipid association can stabilize a particular conformation. The model used here does not include that micelle environment.

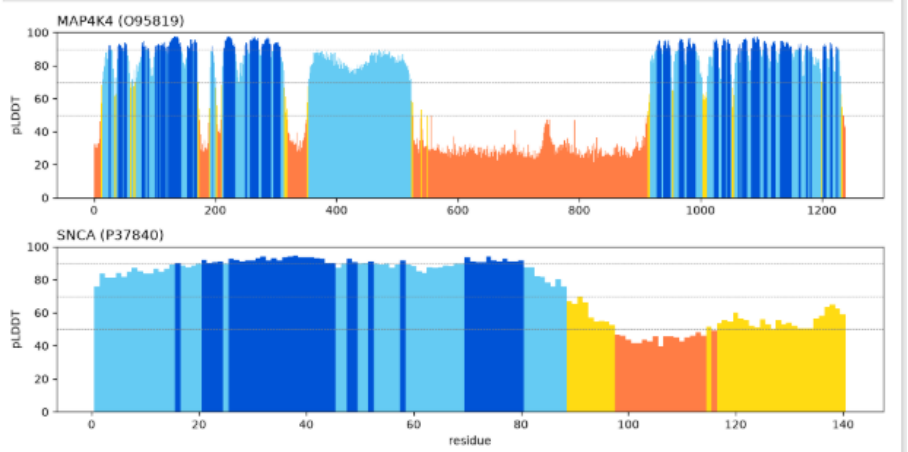
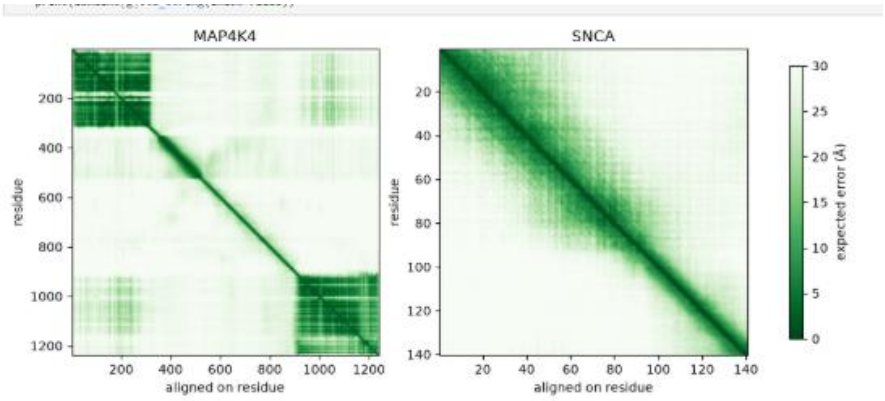


Figure 1. Per-residue pLDDT profiles for MAP4K4 and SNCA. MAP4K4 contains extensive low-confidence regions, while SNCA confidence declines markedly after residue 88.




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MAP4K4: 3 confident domain(s); mean PAE between domains 1 and 2: 28.1 Å
 start  end  confident_residues
    14  313                261
   354  522                169
   917 1231                299

SNCA: 1 confident domain(s); mean PAE between domains 1 and 2: nan Å
 start  end  confident_residues
     1   88                 88
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Figure 2. PAE maps for MAP4K4 and SNCA. The MAP4K4 inter-domain errors indicate uncertainty in relative domain placement; SNCA does not show a clear low-PAE square establishing a rigid domain across residues 1–88.

MAP4K4 compared with 4U43

After correcting the alignment procedure, the MAP4K4 prediction superposed with experimental structure 4U43 at an RMSD of 1.19 Å over 259 matched residues, compared with 20.99 Å from the initial alignment. The 4U43 structure is an inhibitor-bound MAP4K4 kinase-domain structure containing ligand 3D8, determined at 2.18 Å resolution (RCSB Protein Data Bank, n.d.). The low RMSD supports good agreement across the aligned region, but it should not be generalized to the full-length protein because the experimental structure represents a limited construct rather than the entire predicted sequence.

The largest reported deviations were residue 33 (4.4 Å), residue 294 (3.8 Å), residue 311 (3.7 Å), residue 310 (2.8 Å), residue 32 (2.8 Å), and residue 46 (2.7 Å). Deviations around residues 31, 33, 34, and 39 overlap the inhibitor-contact region. These differences may reflect ligand-associated conformational effects, local loop flexibility, or prediction error; the RMSD values alone cannot distinguish among these explanations. Deviations near residues 294 and 310–311 also occur near the end of the aligned region and should be interpreted with care.

SNCA compared with 1XQ8

The whole-structure RMSD between the SNCA prediction and 1XQ8 was 32.45 Å, calculated over the 88 confident residues spanning residues 1–88. Segment-specific comparisons produced lower RMSDs: 1.36 Å for residues 3–37 and 3.77 Å for residues 45–92. Thus, the large whole-structure RMSD is not caused by including the low-confidence C-terminal tail. It primarily reflects differences in the relative arrangement and curvature of the helical regions. The deposited 1XQ8 structure is the average NMR structure of micelle-bound α-synuclein, which adopts two antiparallel helices in that environment (Ulmer et al., 2005). The comparison therefore reflects a combination of differences in helix arrangement and the context-dependent nature of the protein; the RMSD alone cannot determine how much each factor contributes.

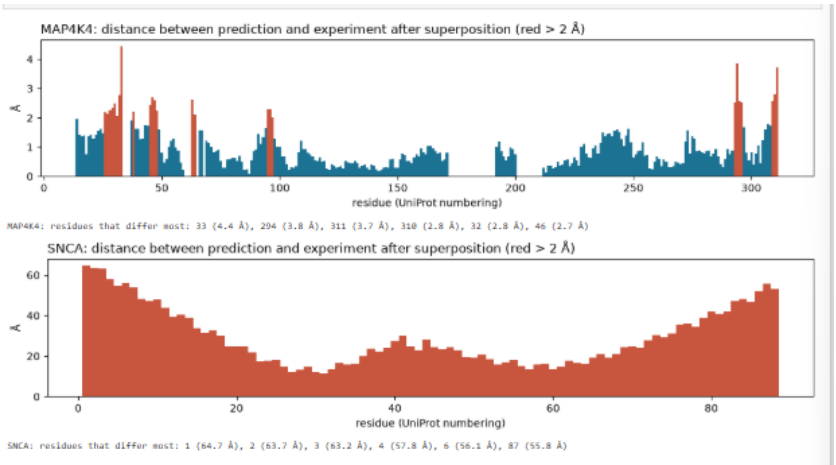


Figure 3. Per-residue deviations after superposition. MAP4K4 deviations are concentrated in particular regions, while the whole-chain SNCA comparison shows large deviations despite lower RMSDs for individual helical segments.

MAP4K4

The highest-ranked MAP4K4 pocket retained for analysis was pocket 60, but it overlapped 0% of the known inhibitor site. This means the highest-ranked retained pocket is not the experimentally established inhibitor-binding site. The closest site-matching pocket was pocket 122, ranked 31st with an fpocket score of 0.002. It was rejected by the confidence-aware assessment because it merged with low-confidence residues 172–191, with pLDDT values of approximately 28–54, and included distant C-terminal residues 993 and 1052–1056. This raises concern that the predicted pocket geometry is influenced by uncertain regions or by the way distant parts of the full-length model are positioned.

The experimental 4U43 structure nevertheless demonstrates that MAP4K4's kinase domain can bind an inhibitor. The predicted model's failure to recover that site cleanly does not prove that MAP4K4 is undruggable. It instead indicates that pocket detection on

this full-length prediction is not sufficient to evaluate the known inhibitor site. The core-site comparison should be used to determine whether the 15 inhibitor-contact residues are locally better supported than pocket 122.

Core-site comparison: The mean pLDDT and mean PAE for the 15 inhibitor-contact residues were **[33 (4.4 Å), 294 (3.8 Å), 311 (3.7 Å), 310 (2.8 Å), 32 (2.8 Å), 46 (2.7 Å]**. Pocket 122 had a mean pLDDT of 67.9 and mean PAE of 11.1 Å. These values should be interpreted together: pLDDT reflects local confidence, whereas PAE estimates positional uncertainty. A lower PAE is not interchangeable with a higher pLDDT, and neither metric alone establishes ligand binding.

SNCA

The highest-scoring retained SNCA pocket was pocket 1, with a score of 0.949. Its lining residues span approximately residues 48–63, with residues spaced at intervals consistent with positions i, i+3, and i+4 along an α-helix. This pattern is compatible with a pocket formed along one face of a helix. However, a high fpocket score does not establish that the site binds a ligand, is selective, or affects SNCA function.

The predicted pocket must be interpreted in the context of SNCA's structural flexibility, the lower confidence of the C-terminal region, and the possibility that membrane-associated states differ from the predicted conformation. A pocket that appears in one predicted state may not persist in other biologically relevant states.

Top-five fpocket results

Table 1. MAP4K4: top five fpocket pockets with confidence filter

Rank	Pocket	Druggability score	Volume (Å³)	Mean pLDDT	PAE within (Å)	Verdict
1	95	0.204	678.4	72.9	11.7	Suspect
2	60	0.103	424.0	72.6	7.9	Keep
3	87	0.074	171.6	66.9	10.4	Reject
4	91	0.035	409.4	87.7	1.8	Keep
5	13	0.027	182.2	84.0	1.6	Keep
31	122	0.002	—	67.9	11.1	Reject; covers 87% of the 3D8 inhibitor site

Table 2. SNCA: top five fpocket pockets with confidence filter

Rank	Pocket	Druggability score	Volume (Å³)	Mean pLDDT	PAE within (Å)	Verdict
1	1	0.949	472.7	89.4	7.4	Keep
2	2	0.630	489.9	52.7	6.5	Reject
3	10	0.497	255.5	87.3	3.9	Keep
4	8	0.155	251.7	87.1	4.6	Keep
5	9	0.092	171.3	55.3	3.5	Reject

Explicit druggability calls

MAP4K4:

The top-ranked MAP4K4 pocket overall was pocket 95, with a druggability score of 0.204, volume of 678.4 Å³, mean pLDDT of 72.9, and within-pocket PAE of 11.7 Å. It was classified as suspect rather than kept because it spans residues 182–1056, creating a pocket across the kinase/C-terminal-domain interface that is affected by uncertain relative domain placement. Pocket 60 was the second-ranked pocket overall and the highest-ranked pocket retained for analysis; its score was 0.103, volume was 424.0 Å³, mean pLDDT was 72.6, and within-pocket PAE was 7.9 Å. However, pocket 60 overlapped 0% of the known inhibitor site, so it should not be confused with the experimentally established binding site.

The closest site-matching pocket was pocket 122, ranked 31st, with a score of 0.002, mean pLDDT of 67.9, and within-pocket PAE of 11.1 Å. It covered 87% of the 3D8 inhibitor site but was rejected because it merged with low-confidence residues 172–191 and included distant C-terminal residues 993 and 1052–1056. The known ligand site is therefore poorly represented by the confidence-filtered pocket ranking, despite being present in the inhibitor-bound experimental structure. The mean pLDDT and PAE for the 15 inhibitor-contact residues are **[72.9, 72.6, 66.9, 87.7, 84.0]** and **[11.7, 7.9, 10.4, 1.8, 1.6] Å**, respectively. Comparing these values with pocket 122 (67.9 and 11.1 Å) helps assess whether the known contact region is locally more reliable than the predicted pocket boundary, while recognizing that pLDDT and PAE measure different aspects of structural confidence.

SNCA:

SNCA pocket 1 was retained, with a druggability score of 0.949, volume of 472.7 Å³, mean pLDDT of 89.4, and within-pocket PAE of 7.4 Å. These confidence values support the local reliability of the predicted pocket geometry, but they do not establish ligandability. Its lining residues, approximately 48–63, lie along one face of a single α-helix, with positions spaced by i, i+3, and i+4. This arrangement may provide a shallow groove, but a single helix face does not necessarily provide sufficient enclosure to surround and stabilize a small molecule. Because enclosure is an important consideration in evaluating pocket druggability (BIOT6900, Module 4 lecture slides, slide 19), the pocket should be treated as a candidate surface feature rather than a validated enclosed binding site. Its persistence and accessibility also need to be assessed in biologically relevant SNCA conformations.

MAP4K4 ranked fourth in Module 2 and first in Module 3, with p_sm = 0.665. Its predicted domains can be interpreted locally where confidence is adequate, and the inhibitor-bound 4U43 structure confirms that the kinase domain can bind a small molecule. However, the full-length model has uncertain inter-domain placement, and the pocket analysis did not robustly recover the known inhibitor site. MAP4K4 should therefore be assessed using ligand-informed analysis of the kinase domain rather than relying on the highest-scoring pocket in the full-length prediction.

SNCA ranked third in Module 2, with a GWAS −log10 p = 169.4; its Module 3 values were p_sm = 0.023 and p_ab = 0.001. These results support biological prioritization, but they do not validate pocket 1 as a stable small-molecule binding site. Although pocket 1 has a mean pLDDT of 89.4 and within-pocket PAE of 7.4 Å, its one-helix-face geometry raises a question about enclosure, and the predicted model lacks the lipid environment relevant to SNCA structure. Both targets remain worthy of consideration, but the structural evidence supports different next steps: ligand-informed kinase-site evaluation for MAP4K4 and context-aware pocket validation for SNCA.

Main limitations and how could they be addressed:

1. One snapshot. A predicted structure represents one model rather than the full conformational ensemble. For MAP4K4, the relative arrangement of domains and the geometry of loops near the inhibitor-contact region may vary. For SNCA, helix curvature and arrangement can change with its molecular environment. A useful next step would be to compare additional structures or use molecular-dynamics simulations to evaluate which features persist across conformations.

2. Missing pieces. The predicted MAP4K4 structure lacks its bound inhibitor and does not include any potential partner proteins that could influence the kinase-domain environment. The SNCA model lacks the lipid or micelle environment that stabilizes its experimentally observed helical conformation. These absent components can affect pocket geometry and accessibility. For MAP4K4, pocket analysis should be repeated on the inhibitor-bound 4U43 structure and compared with an appropriate ligand-free structure. For SNCA, the pocket should be assessed in a relevant lipid-associated structural context before concluding that it represents a persistent binding site.

3. No membrane. SNCA's helical structure is sensitive to lipid environment. The two antiparallel helices in the micelle-bound 1XQ8 average structure demonstrate that its conformation depends on context (Ulmer et al., 2005). A next step would be to evaluate pocket persistence under membrane-mimetic conditions and compare the result with the predicted structure.

4. Cryptic pockets. A binding site may appear or change shape when a protein adopts a different conformation or binds a ligand. MAP4K4 illustrates this limitation because the highest-ranked pocket, pocket 95, is suspect at a kinase/C-terminal-domain interface, while the site-matching pocket 122 is low-ranked and rejected by the confidence filter. Comparing pockets in inhibitor-bound and ligand-free MAP4K4 structures would test whether the known site depends on ligand-associated structural changes. A biochemical kinase-inhibition assay could then determine whether binding translates into functional inhibition.

Conclusion

MAP4K4 remains a candidate for ligand-informed kinase-site evaluation, but its predicted full-length pocket results do not reliably recover the known inhibitor site. SNCA remains biologically relevant, although its context-dependent structure means the predicted pocket requires independent validation. In both cases, structural confidence and pocket detection should guide experiments rather than be treated as proof of druggability.

Step 8 · Anchor check: p53

The predicted p53 structure closely matches the experimental reference structure 1TUP chain B, with an RMSD of 0.68 Å across 192 confident residues (pLDDT ≥ 70), compared with 0.73 Å across all 194 residues, supporting good overall structural agreement despite localized deviations. The hotspot residues C176, H179, C238, and C242 coordinate zinc, with their reported distances to Zn ranging from 2.1 to 2.3 Å, whereas R248 and R273 directly contact DNA at 3.4 and 2.5 Å, respectively. These structural observations provide a functional basis for interpreting the Module 3 sequence-based variant scores: substitutions at zinc-ligand positions may disrupt metal coordination and protein stability, while DNA-contact substitutions may impair p53–DNA interactions. Y220 is an important exception because it lies far from both zinc and DNA and was among the less extreme hotspots, yet the Y220C mutation can create a druggable crevice, illustrating how a variant can affect structural properties without directly disrupting either interaction. Correctly interpreting this p53 anchor therefore helps establish a structural framework for evaluating our own variant targets, while reminding us that sequence-based scores and predicted structures provide complementary evidence rather than definitive proof of functional effects.

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